

Electronics Repair Shop Management System

A future-proof business management system for an electronics repair shop. The system is designed as a modular business platform where repair management is the core, supported by accounting, inventory, purchasing, customers, employees, sales, reporting, and future integrations.

1. Overall System

The system should not be built as only a repair-ticket application. It should operate as a connected business management platform. A repair can affect inventory, sales, accounting, notifications, and reports automatically.

Example flow: Customer brings in device → repair ticket created → technician diagnoses it → parts are consumed → inventory decreases → invoice generated → payment recorded → accounting entries generated → reports updated.

2. Core Modules

Module	Purpose
Dashboard	Today's sales, repair revenue, pending repairs, overdue repairs, unpaid invoices, low stock, expenses, cash balance, and profit.
Customer Management	Customer profiles, contact information, devices, repair history, purchases, invoices, payments, balances, and notes.
Repair Management	Repair tickets, device information, diagnosis, status workflow, estimates, approvals, technician assignment, notes, photos, and history.
Device Tracking	Device type, brand, model, serial number, IMEI, condition, accessories, photos, customer ownership, and repair history.
Inventory	Parts and products, stock levels, suppliers, costs, selling prices, stock movements, locations, and low-stock alerts.
Purchasing	Suppliers, purchase orders, deliveries, supplier bills, payments, and purchase history.
POS / Sales	Sales of chargers, cables, batteries, accessories, replacement parts, used devices, and other products.
Accounting	Chart of accounts, journal entries, general ledger, accounts receivable, accounts payable, expenses, and financial reports.
Employees	Employees, roles, permissions, branches, assignments, commissions, activity, and performance.
Reporting	Sales, repairs, inventory, financial, employee, and business-performance reports.
Notifications	Repair updates, customer approvals, pickup notices, payment reminders, and inventory alerts.
Audit Log	Record important changes to repairs, inventory, prices, payments, users, and accounting records.

3. Repair Management

Repair management should be the heart of the application and should support a configurable workflow.

- Receive device
- Record device condition and accessories
- Record customer-reported problem
- Perform diagnosis
- Create estimate
- Wait for customer approval
- Approve repair
- Wait for parts if necessary
- Repair device
- Test device
- Mark ready for pickup
- Complete transaction

4. Repair Ticket

Each repair ticket should contain the customer, device, problem, condition, accessories, technician, status, estimated cost, approval status, parts, payments, notes, photos, and warranty information.

Field	Example
Repair	#10452
Customer	John Doe
Device	Dell Inspiron 15
Serial Number	ABC123456
Problem	Laptop will not turn on
Condition	Minor scratches
Accessories	Charger, laptop bag
Technician	Mike
Status	Diagnosis
Estimated Cost	■4,500
Customer Approval	Pending

5. Device Tracking

- Device type
- Brand and model
- Serial number
- IMEI where applicable
- Color and identifying details
- Condition
- Accessories received
- Photos
- Customer ownership
- Previous repair history

6. Inventory Management

- SKU, name, category, brand, model, supplier, cost, selling price, quantity, minimum stock, and storage location.
- Track every stock change as an inventory movement.
- Support purchases, repairs, sales, returns, damaged items, and manual adjustments.
- Automatically deduct parts when used in repairs.
- Provide low-stock alerts and future purchase suggestions.
- Support serial and batch tracking where needed.

7. Purchasing

- Supplier profiles
- Supplier products
- Purchase orders
- Receiving

- Supplier bills
- Supplier payments
- Purchase history
- Automatic inventory updates

8. POS / Sales

- Sell parts, accessories, chargers, cables, batteries, used devices, and other products.
- Automatically reduce inventory when a sale is completed.
- Generate invoices and receipts.
- Support discounts, payments, returns, and customer-linked sales.

9. Accounting

Accounting should be designed around a proper ledger instead of simply storing sales and expenses.

- Chart of accounts
- Journal entries
- General ledger
- Accounts receivable
- Accounts payable
- Cash and bank accounts
- Sales revenue
- Repair revenue
- Inventory
- Cost of goods sold
- Labor and operating expenses
- Financial reporting

Business transactions should automatically create the appropriate accounting records. This prevents staff from manually entering the same transaction multiple times.

10. Automatic Accounting Flow

Business Transaction → Accounting Event → Journal Entry → General Ledger → Profit & Loss / Balance Sheet / Cash Flow / Other Reports

Example: when a customer pays for a repair, the system records the payment and associated revenue. When a part is consumed, inventory decreases and the corresponding cost is recorded.

11. Expenses

- Rent
- Electricity
- Internet
- Salaries
- Supplies
- Transportation
- Equipment
- Software
- Advertising
- Miscellaneous expenses
- Recurring expenses

12. Employee Management

- Employee profiles
- Roles and permissions
- Branch assignment
- Technician assignments
- Repair activity

- Sales activity
- Performance information
- Commission support for future use

13. Role-Based Permissions

Role	Typical Permissions
Owner	Full access to business operations, accounting, reporting, settings, and audit history.
Admin / Manager	Operations, repairs, inventory, purchasing, employees, and selected reports.
Cashier	Sales, invoices, payments, receipts, and customer transactions.
Technician	Assigned repairs, diagnosis, notes, parts used, and repair status.
Inventory Staff	Products, stock movements, receiving, and purchasing.

14. Audit Log

Important business changes should never disappear without a trace. The system should record who changed what, when it happened, and the relevant before-and-after values.

- Repair status changes
- Inventory adjustments
- Price changes
- Payment changes
- Deleted or cancelled transactions
- Permission changes
- Accounting changes

15. Reporting

Category	Reports
Sales	Daily, weekly, monthly sales; sales by employee, product, and category.
Repairs	Repairs received, completed, pending, overdue, average repair time, failed repairs, warranty returns.
Inventory	Current stock, low stock, stock valuation, inventory movements, dead stock, turnover.
Financial	Profit & Loss, Balance Sheet, Cash Flow, revenue, expenses, COGS, receivables, payables.
Employees	Repairs completed, revenue generated, workload, completion time, performance.
Business	Revenue trends, most repaired devices, most profitable services, and operational performance.

16. Notifications

- Repair received
- Estimate ready
- Customer approval required
- Repair delayed
- Repair completed
- Ready for pickup
- Payment reminder

- Low-stock alert

The notification system should be configurable so future channels such as SMS, email, push notifications, or customer portals can be added without changing repair logic.

17. Customer Portal

- View repair status
- Approve or decline estimates
- View invoices
- View payment status
- Download receipts
- View repair history
- Make online payments in the future

18. Multi-Branch Support

Even if the first shop has only one location, the system should be designed around a company and branch structure.

- Branches have their own inventory
- Employees can belong to branches
- Repairs can belong to branches
- Sales can belong to branches
- Reports can be filtered by branch
- Inventory can eventually be transferred between branches

19. Configurable Settings

- Repair statuses
- Repair categories
- Product categories
- Payment methods
- Tax configuration
- Invoice numbering
- Warranty periods
- Notification templates
- User roles and permissions
- Branches
- Business information
- Other business rules

20. Future Modules

- Customer portal
- Supplier portal
- Online booking
- Multi-branch management
- Mobile technician application
- Warranty management
- E-commerce
- Service contracts
- Subscription billing
- Advanced analytics
- AI business assistant
- Additional industry-specific modules

21. Recommended Architecture

The application should use a modular backend rather than immediately splitting every feature into a separate microservice.

- Web and mobile clients
- API / GraphQL layer
- Authentication and authorization
- Business modules for CRM, repairs, sales, inventory, purchasing, and accounting
- Event/domain layer for cross-module business events
- Notification and audit services
- Reporting layer
- Relational database such as PostgreSQL or SQL Server

22. Key Architectural Principle

Build around business transactions rather than screens. For example:

Repair Completed → Inventory Consumed → Revenue Recorded → Accounting Updated → Notification Triggered → Reports Updated

This makes it possible to add future modules such as customer portals, loyalty programs, analytics, mobile applications, and integrations without rewriting the repair system.

23. Suggested MVP Roadmap

Phase 1 — Core — Authentication, users, roles, customers, devices, repair tickets, repair workflow, technicians, dashboard, audit logs.

Phase 2 — Inventory — Products, parts, categories, suppliers, stock movements, purchase orders, low-stock alerts, inventory valuation.

Phase 3 — Sales — POS, invoices, payments, receipts, discounts, returns.

Phase 4 — Accounting — Chart of accounts, journal entries, general ledger, receivables, payables, expenses, Profit & Loss, Balance Sheet, Cash Flow.

Phase 5 — Automation — Automatic inventory deductions, automatic accounting, notifications, recurring expenses, scheduled reports.

Phase 6 — Expansion — Customer portal, supplier portal, multi-branch, mobile technician app, online booking, warranty management, advanced analytics.

24. Final Direction

The recommended approach is to build an electronics repair shop as the first vertical of a broader business-management platform. The repair module remains the specialized core, while customers, inventory, accounting, reporting, authentication, audit logs, notifications, settings, and integrations form reusable platform capabilities.

This approach allows the system to start small while remaining capable of supporting additional features, branches, business types, and integrations in the future.